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Toward Precision Medicine: Explaining Mortality Risk Predictions in Heart Failure Patients

A Health Data Analytics Project Combining Statistical Analysis,
Machine Learning, and Explainable AI



Heart Failure Mortality Risk Analysis

Project Report

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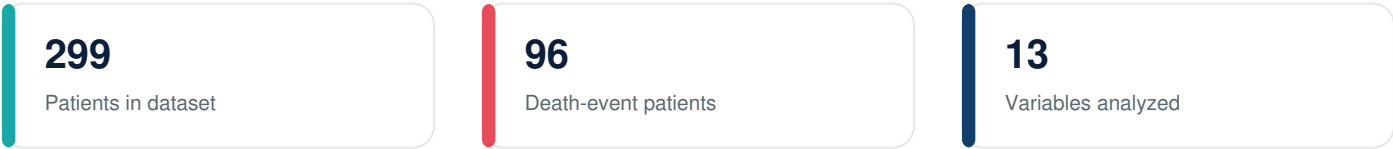
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1. Executive Summary

A concise overview of the precision medicine storyline and key project outputs.



This project investigates mortality risk among heart failure patients using a workflow that combines exploratory data analysis, statistical testing, machine learning, and SHAP-based explainable AI. The central idea is that precision medicine is not only about predicting whether a patient is high risk. It is also about understanding why a model sees that patient as high risk and which clinical features contributed to the prediction.

The dataset contains 299 heart failure patient records and a binary outcome called DEATH_EVENT. In this project, DEATH_EVENT = 0 means no death event was observed during follow-up, while DEATH_EVENT = 1 means a death event was observed. The data had no missing values and no duplicate rows.

The workflow moved from dataset understanding to EDA, statistical testing, machine learning evaluation, and patient-level explanation. Models were evaluated with healthcare-relevant metrics, including precision, recall, F1-score, confusion matrix, and predicted probability. SHAP was then used to explain both global patterns and individual patient predictions.

Our results are for educational purposes only. The models are not clinical decision systems, and the explanations should be interpreted as descriptions of model behavior rather than proof of medical causality.

Why precision medicine matters

Patients with the same disease can have different risk profiles because of differences in age, heart function, kidney function, blood chemistry, comorbidities, and lifestyle-related variables. Predictive models can support precision medicine only when they are evaluated responsibly and explained clearly.

Variable	Simple clinical meaning
Ejection fraction	How well the heart pumps blood
Serum creatinine	Marker related to kidney function
Serum sodium	Blood chemistry balance
Age / comorbidities	Patient profile features that may contribute to risk patterns

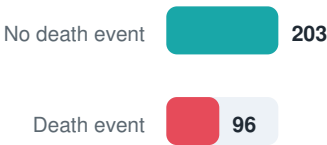
2. Dataset Overview and Project Workflow

From raw clinical records to explainable model predictions.

Each row represents one heart failure patient. The variables include demographic information, clinical measurements, comorbidity indicators, follow-up time, and the death-event outcome. The project uses cautious language such as "associated with," "contributed to predicted risk," and "model-based explanation."

Dataset item	Project value
Patients	299
Variables	13
Target	DEATH_EVENT
No death event	203 patients
Death event	96 patients
Missing values / duplicates	None / none

Death-event distribution

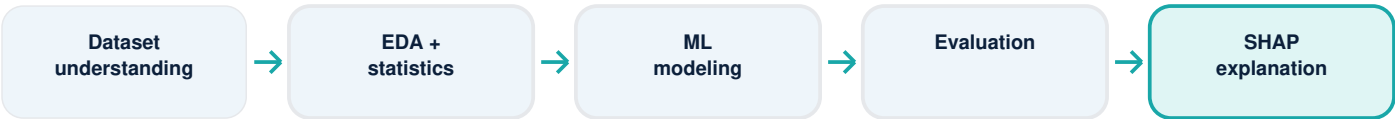


Target: DEATH_EVENT (0 = no death event ; 1 = death event)

Variable groups

Group	Variables
Demographic	age, sex
Clinical measurements	creatinine_phosphokinase, ejection_fraction, platelets, serum_creatinine, serum_sodium
Comorbidity and lifestyle	anaemia, diabetes, high_blood_pressure, smoking
Follow-up and outcome	time, DEATH_EVENT

Project workflow



Important modeling choice: time was excluded from the classification feature set because follow-up time is part of the survival outcome process. This helps reduce leakage risk and motivates survival analysis as the main future direction.



3. Exploratory and Statistical Analysis

Understanding patient characteristics before predictive modeling.

The exploratory analysis checked dataset shape, column types, missing values, duplicates, target distribution, numerical variable distributions, and categorical comparisons. Histograms and boxplots were used to inspect skewness and outliers. Crosstabs helped compare death-event proportions across categorical variables.

Several clinical variables showed distributional differences between death-event groups. Patients with a death event tended to be older, had lower ejection fraction, higher serum creatinine, and lower serum sodium. These patterns support model interpretation, but they do not prove causality.

Variable	Test / comparison	p-value	Interpretation
Age	Mann-Whitney U, death group greater	8.34e-05	Death-event group tended to be older
Serum creatinine	Mann-Whitney U, death group greater	7.90e-11	Death-event group tended to have higher values
Serum sodium	Mann-Whitney U, death group lower	0.000146	Death-event group tended to have lower values
Ejection fraction	Mann-Whitney U, death group lower	3.68e-07	Death-event group tended to have lower values
Categorical variables	Crosstabs / chi-square where applicable		Useful for group-level association checks

Statistical testing supports evidence-based interpretation before modeling. However, this is an observational dataset. A low p-value means the variable distribution differed between groups under the test assumptions; it does not prove that the variable caused mortality.



4. Machine Learning Modeling and Evaluation

Predicting DEATH_EVENT while evaluating the minority death-event class responsibly.

The machine learning task was framed as binary classification: predict whether a patient experienced a death event during follow-up. The data was split into training and test sets using stratification, and preprocessing pipelines were used to avoid leakage. Because the death-event class is the minority class, evaluation focused on precision, recall, F1-score, confusion matrix, and predicted probability rather than accuracy alone.

Model	Death-event precision	Death-event recall	Death-event F1	Portfolio interpretation
Tuned Logistic Regression	0.61	0.74	0.67	Best balance of interpretability and death-event detection
Tuned Random Forest	0.52	0.63	0.57	Moderate performance after tuning
Tuned Gradient Boosting	0.58	0.74	0.65	Comparable recall; useful nonlinear SHAP explanation
Tuned SVM	0.46	0.95	0.62	Very high recall, lower precision
Tuned KNN	0.45	0.26	0.33	Weak death-event detection

Healthcare meaning of errors

False negative: the model predicted lower risk for a patient who actually experienced the death event.

False positive: the model predicted higher risk for a patient who did not experience the death event.

Predicted probability: the model estimate of death-event risk for a patient, not a diagnosis.

Selected confusion matrices

Model	TN	FP	FN	TP
Logistic Regression	32	9	5	14
Gradient Boosting	31	10	5	14

Rows summarize test-set class counts for selected models.

Models selected for SHAP

Tuned Logistic Regression was selected as a clinically readable baseline. Tuned Gradient Boosting was selected because it can capture nonlinear relationships and feature interactions. Comparing both explanations helps show whether similar clinical signals appear across different modeling assumptions.

5. Explainable AI and Global SHAP Interpretation

SHAP explains model predictions; it does not prove clinical causality.

SHAP helps answer: "Why did the model make this prediction for this patient?" Global SHAP summarizes feature influence across test patients, while patient-level SHAP explains one patient at a time. Positive SHAP values push the model toward the death-event class; negative values push it away.

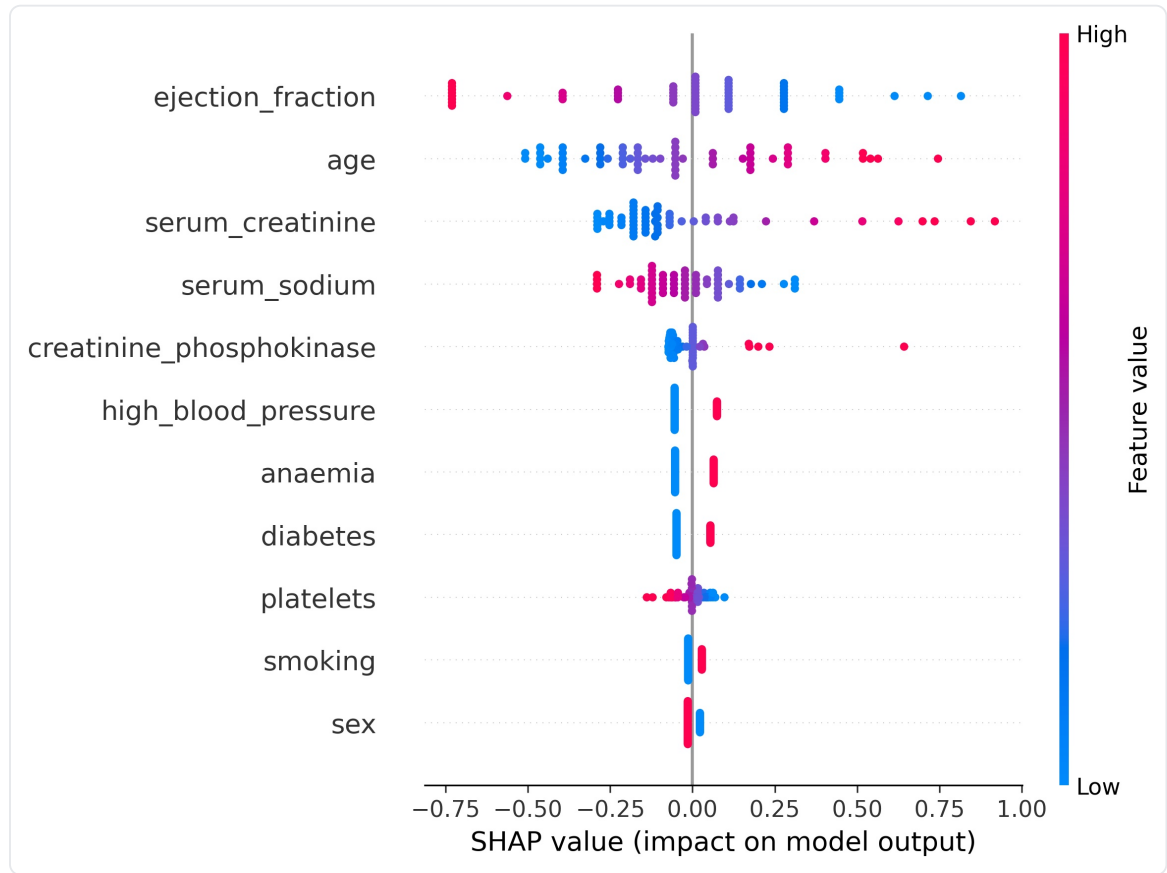


Figure 1. Logistic Regression SHAP beeswarm plot. Features on the right push predictions toward higher death-event risk; features on the left push predictions lower. Color reflects feature value.

Model	Top features by mean absolute SHAP
Logistic Regression	ejection_fraction (0.274), age (0.265), serum_creatinine (0.216), serum_sodium (0.117)
Gradient Boosting	serum_creatinine (0.126), ejection_fraction (0.087), age (0.059), platelets (0.047)

Both selected models highlighted ejection fraction, serum creatinine, and age as important contributors to predicted mortality risk. This agreement is useful because the two models have different assumptions: Logistic Regression is linear, while Gradient Boosting can capture nonlinear patterns.

6. Patient-Level SHAP Explanation: True Positive

A correctly identified death-event patient.

Patient 110 had an actual death-event outcome. The Gradient Boosting model predicted the death-event class with a predicted death-event probability of 53.6%.

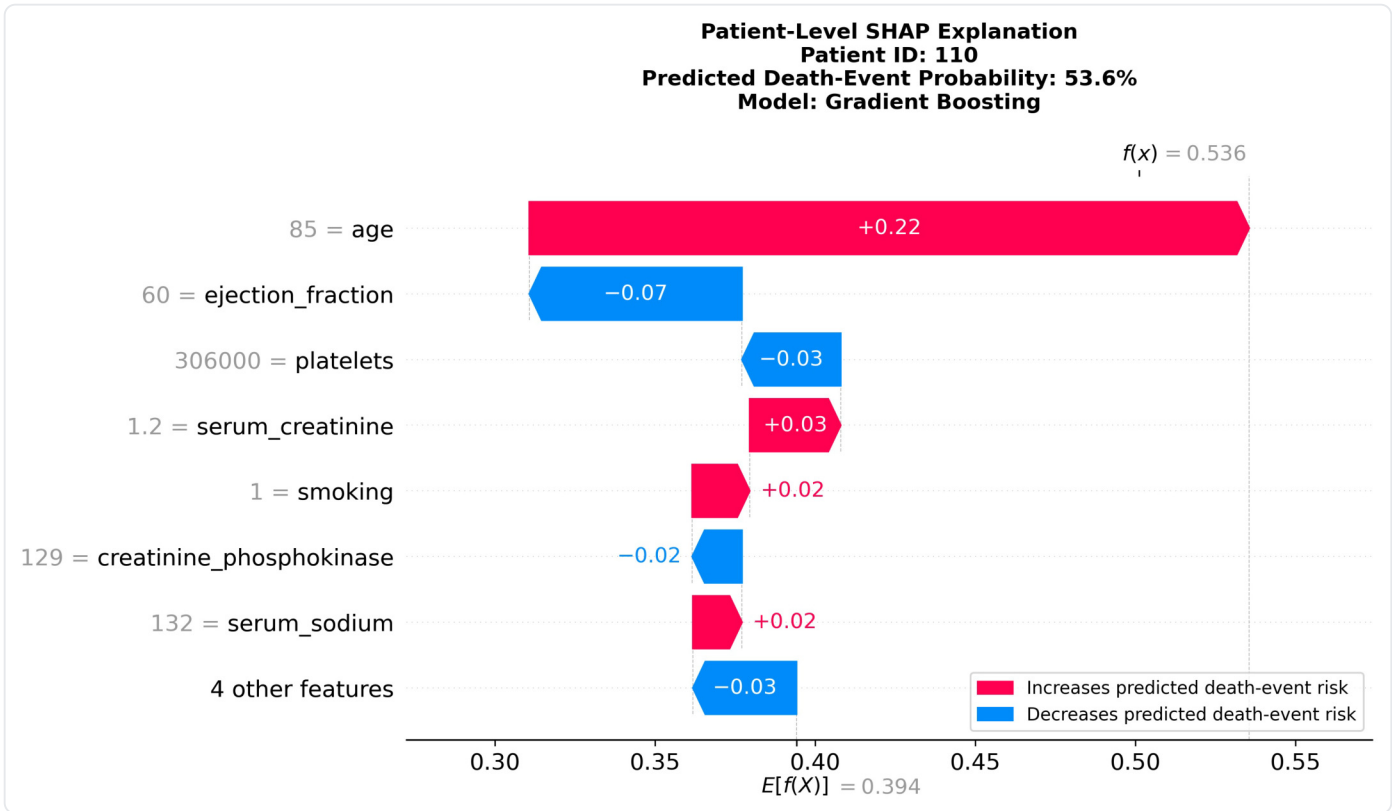


Figure 2. Patient-level SHAP waterfall plot for Patient 110. Red features increased predicted death-event risk; blue features decreased predicted risk.

Interpretation

Age, serum creatinine, smoking, and serum sodium pushed the model toward the death-event class. Ejection fraction, platelets, creatinine phosphokinase, diabetes status, and other features pushed the prediction lower. This interpretation is model-based and should not be read as evidence that any feature caused the event.

7. Patient-Level SHAP Explanation: True Negative

A correctly identified lower-risk patient.

Patient 149 had an actual no-death-event outcome. The Gradient Boosting model predicted no death event with a predicted death-event probability of 17.6%.

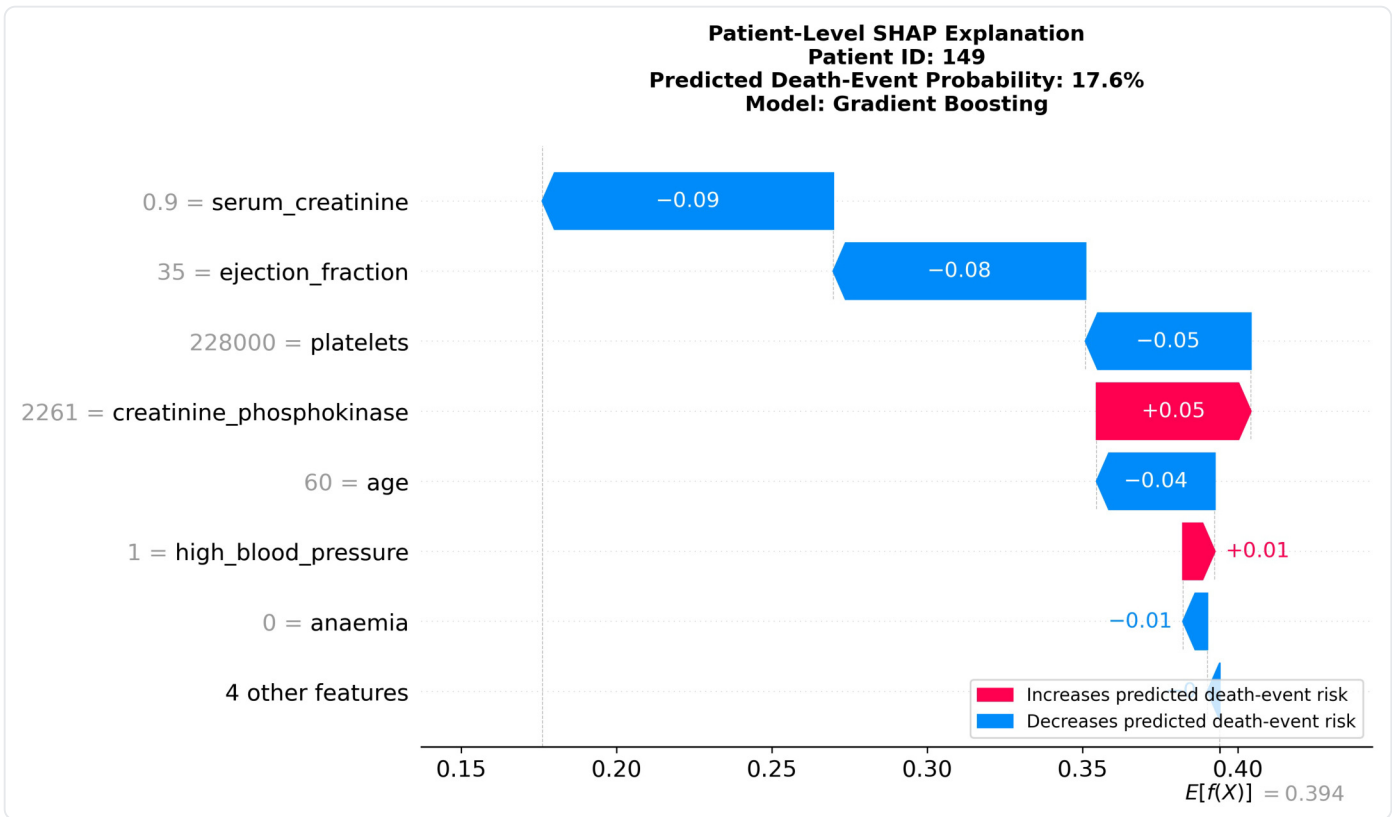


Figure 3. Patient-level SHAP waterfall plot for Patient 149. Most high-impact features moved the prediction away from the death-event class.

Interpretation

Serum creatinine, ejection fraction, platelets, age, anaemia status, and smoking status pushed the prediction away from the death-event class. Creatinine phosphokinase and high blood pressure pushed the prediction slightly higher. This case shows how SHAP can explain a lower-risk model prediction for an individual patient.

8. Patient-Level SHAP Explanation: False Positive

A case where the model overestimated death-event risk.

Patient 190 had an actual no-death-event outcome, but the Gradient Boosting model predicted the death-event class with a predicted death-event probability of 86.4%.

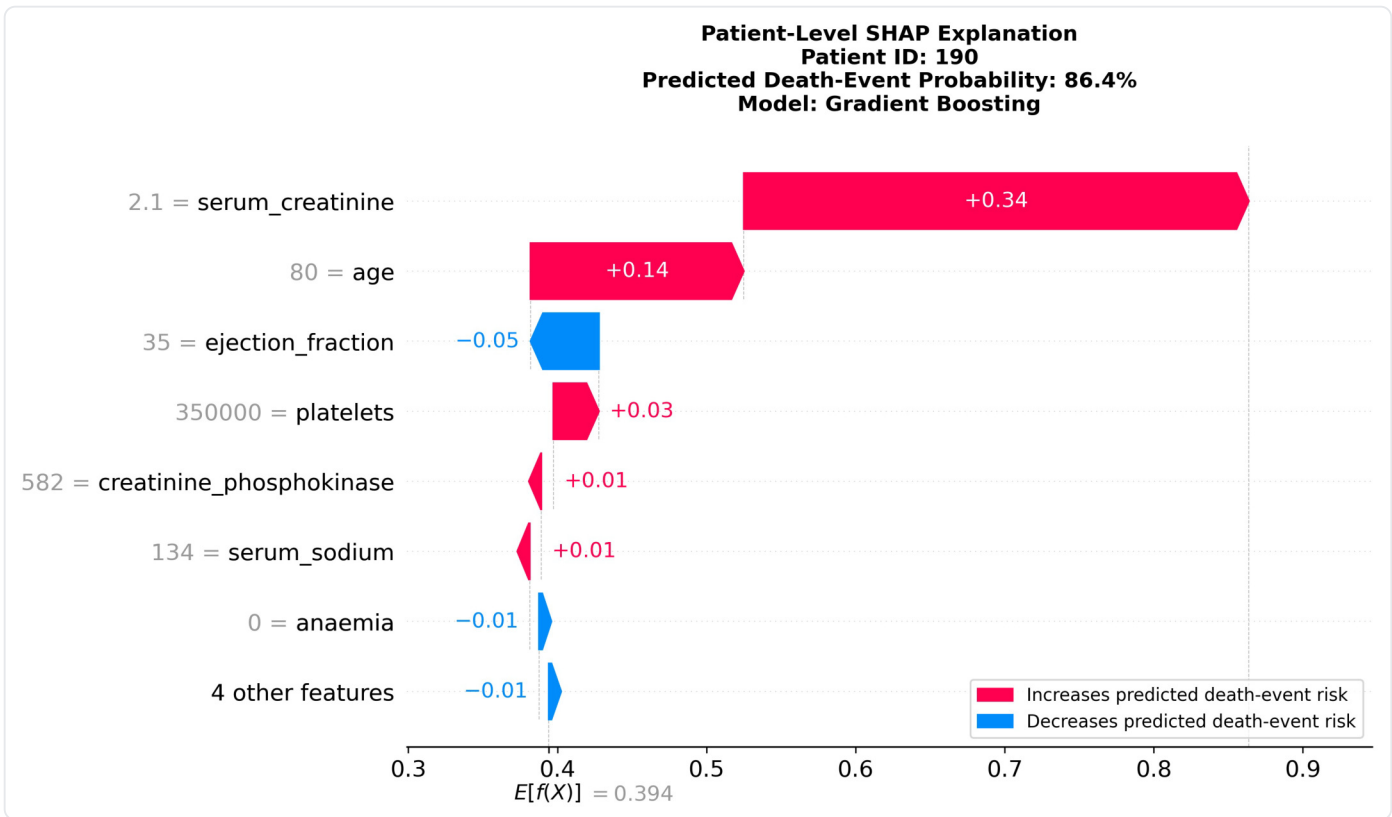


Figure 4. Patient-level SHAP waterfall plot for Patient 190. This error analysis shows which features may have led the model to overestimate risk.

Interpretation

Serum creatinine and age strongly increased the predicted risk for this patient. Platelets, creatinine phosphokinase, and serum sodium also pushed risk slightly higher. Ejection fraction, non-anaemic status, and absence of high blood pressure pushed risk lower. No false negative example was found in the selected Gradient Boosting test results.



9. Discussion, Limitations, Future Direction, and Conclusion

Connecting the analytical workflow back to precision medicine.

The value of this project is not only identifying one best model. The value is showing a responsible workflow where patient data can be used to understand risk patterns, compare clinical profiles, evaluate predictive performance, and explain why individual predictions were made. This is the practical connection to precision medicine: patients can have similar diagnoses but different model-based risk explanations.

Limitation	Why it matters
Small dataset size	Only 299 patients, which limits model stability and generalizability.
No external validation	Performance may not transfer to other hospitals, populations, or clinical settings.
Observational and non-causal	Associations and SHAP explanations do not prove clinical causality.

Future direction: survival analysis

The classification models predict whether a death event occurred. Survival analysis can answer a richer question: "How does risk evolve over time?" This is especially appropriate because the dataset includes both time and DEATH_EVENT. Our future work will include Kaplan-Meier curves, log-rank tests, Cox proportional hazards regression, hazard ratio interpretation, proportional hazards checks, concordance index evaluation, and comparison with machine learning classification.

Conclusion

This project demonstrates how health data analytics can combine clinical reasoning, statistical analysis, machine learning, and explainable AI to support the broader vision of precision medicine. The SHAP explanations move the work beyond black-box prediction by showing which patient-level features contributed to higher or lower predicted mortality risk. Our results should be interpreted as exploratory and educational, not as a clinical decision system. Real-world use would require external validation, clinical expert review, fairness assessment, and prospective testing.